WORKING PROTOTYPE / LIVE DEMO

Real-Time Driver Drowsiness Detection

Computer vision and facial region analysis for low-cost, immediate fatigue alerting

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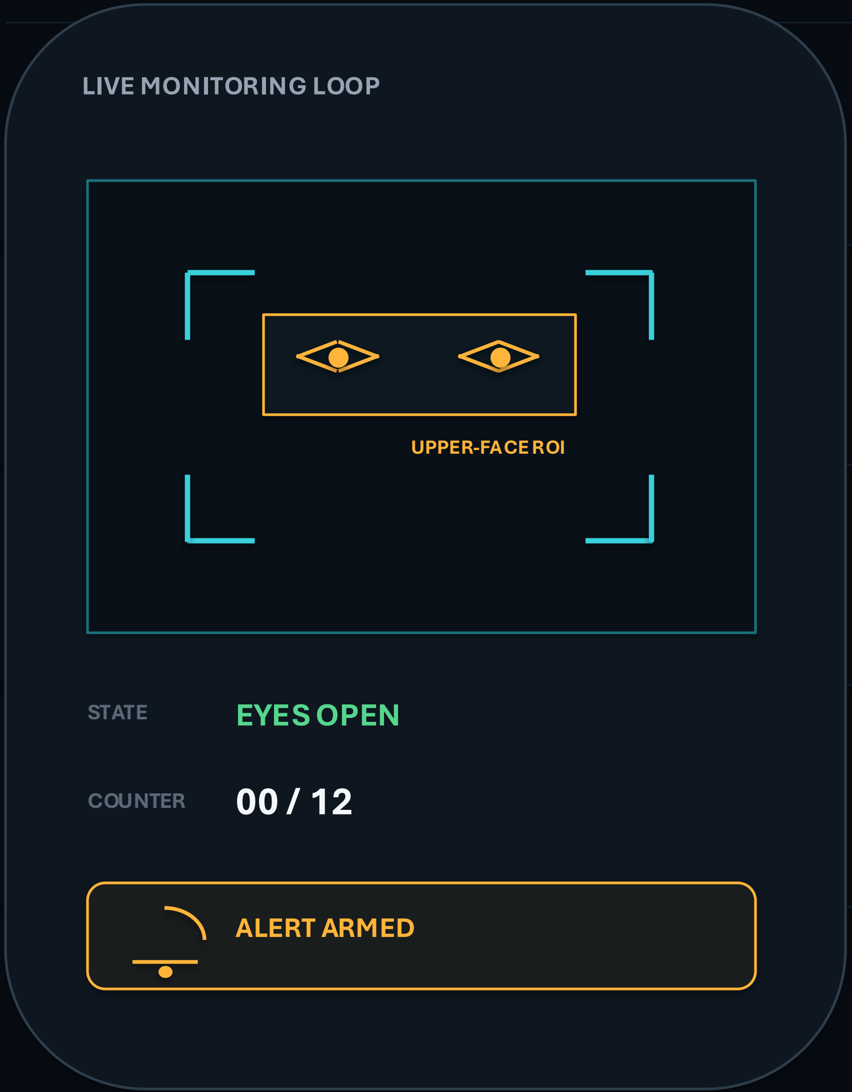
PYTHON 3 / OPENCV / HAAR CASCADES / WINSOUND



01 Why this matters

Drowsiness is dangerous because the driver can lose awareness before realizing performance has degraded.

01

ROAD SAFETY

Detect visible fatigue cues before a microsleep becomes a collision risk.

02

HARDWARE ACCESSIBILITY

Use a standard camera and CPU-based vision instead of specialized biometric sensors.

03

IMMEDIATE ALERTING

Convert sustained eye-closure evidence into an audible and visible warning in real time.

02 The gap

Existing fatigue-monitoring systems are often too expensive, too computationally heavy, or too inaccessible for everyday deployment.

01

SPECIALIZED SENSORS

Infrared cameras, steering sensors, wearables, and commercial driver-monitoring units increase system cost.

02

EDGE COMPUTE LOAD

Complex deep-learning pipelines can introduce latency or require hardware unavailable in low-cost vehicles.

03

ACCESSIBLE REAL-TIME TOOLS

Many prototypes do not provide a simple, reproducible webcam-to-alert workflow for immediate use.

RESEARCH QUESTION

Can a lightweight classical computer-vision pipeline provide practical real-time drowsiness warning on a standard laptop?

03 Research contribution

The contribution is not a new facial detector; it is a deliberately lightweight integration of spatial detection, temporal evidence, and immediate feedback.

01 REAL-TIME LIGHTWEIGHT DETECTION PIPELINE

- Python 3 + OpenCV
- Haar cascade face and eye detection
- Upper-face ROI reduces search area
- Temporal frame counter suppresses single-frame noise

02 LOW-LATENCY MULTI-MODAL ALERT SYSTEM

- On-screen bounding boxes and state text
- Closed-eye frame count shown live
- Windows auditory warning via WinSound
- Threshold-driven alert at 12 consecutive frames

04 Algorithm & methodology

01

FRAME ACQUISITION & GRAY-SCALING

Capture webcam frames and convert BGR input to grayscale for faster cascade evaluation.

02

FACE BOX & UPPER-FACE ROI

Detect the face, then restrict eye search to the upper facial region to reduce false detections.

03

EYE DETECTION & TEMPORAL COUNTER

If eyes are not detected, increment a consecutive closed-frame counter; otherwise reset it.

04

AUDIO / VISUAL ALERT TRIGGER

At counter >= 12, display a warning and trigger an audible alarm until an open-eye state returns.

WHY TEMPORAL COUNTING? Temporal logic converts unstable per-frame detections into a more meaningful persistence signal.

05 Recognition pipeline

FROM RAW PIXELS TO AN ACTIONABLE WARNING STATE

01
CAMERA FEED
BGR frames

02
GRAYSCALE CONVERSION
1-channel image

03
FACE DETECTION
Bounding box

04
EYE CASCADE ROI
Upper-face search

05
CLOSED FRAME COUNTER
Threshold = 12

06
AUDIO / VISUAL OUTPUT
Warning state

TEMPORAL TRACE

1 2 3 4 5 6 7 8 9 10 11 12 ALERT

Decision rule: sustained absence of detected eyes is treated as a drowsiness signal— not a medical diagnosis.

06 Live demo

01	LIVE INPUT	Laptop webcam feed at real-time frame rate
02	VISIBLE STATE	Face and eye bounding boxes rendered on each frame
03	TEMPORAL EVIDENCE	Closed-frame counter displayed continuously
04	INTERVENTION	Audio alert activates when the counter reaches 12

RESPONSE LATENCY Time from sustained eye closure to warning activation

LIGHTING RESILIENCE Qualitative stability under normal, dim, and side-lit conditions

Demo scope: validate the end-to-end interaction loop, not clinical fatigue estimation or production-grade vehicle certification.

07 Real-world deployment

01

COMMERCIAL FLEET MANAGEMENT

Low-cost driver monitoring for delivery, logistics, and long-distance operations.

02

PUBLIC TRANSIT MONITORING

A supplemental alert layer for bus, shuttle, and municipal transport drivers.

03

PERSONAL MOBILITY VEHICLES

Laptop, embedded camera, or compact onboard implementations for private vehicles.

04

INTEGRATED DASHCAM APPS

Software extension for camera systems that already capture the driver-facing view.

DEPLOYMENT PATH

WEBCAM PROTOTYPE — IN-VEHICLE CAMERA — EDGE OPTIMIZATION — FIELD VALIDATION

08 Research challenges

01	VARIABLE AMBIENT LIGHTING	Haar features can become unstable under strong shadows, low light, glare, or rapid illumination changes.
02	EXTREME HEAD POSE	Large yaw, pitch, or partial face occlusion can prevent reliable eye-region detection.
03	EYE-SHAPE FALSE POSITIVES	A fixed cascade and threshold may behave unevenly across individual eye shapes, including smaller or narrower eye appearances.
04	WEBCAM FPS LIMITS	Low frame rate changes the real time represented by 12 frames and can delay or destabilize alert timing.

MITIGATION **calibration, adaptive timing, stronger landmark models, infrared support, and validation across diverse users and conditions.**

09 Conclusion

A LIGHTWEIGHT VISION PIPELINE CAN TURN SUSTAINED EYE-CLOSURE EVIDENCE INTO AN IMMEDIATE DRIVER WARNING.

01 A standard camera can support a practical real-time fatigue warning prototype.

02 ROI restriction and Haar cascades keep the pipeline computationally lightweight.

03 A 12-frame temporal threshold converts noisy detections into an actionable alert state.

04 The next research step is systematic testing across users, lighting, poses, and camera frame rates.

Q&A

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PROJECT STATUS WORKING PROTOTYPE

OPEN FOR QUESTIONS, TESTING FEEDBACK, AND RESEARCH COLLABORATION